

UFC Data Dictionary Explainer

What every stored field means and why it is used

Companion PDF generated from docs/data_dictionary.md. It covers warehouse tables, feature tables, prediction tables, dashboard views, enum values, feature versions, and table relationships.

PRIMARY PURPOSE	Interview-ready explanation of all stored project data
SOURCE DOCUMENT	docs/data_dictionary.md
CORE IDEA	Raw UFC data becomes leakage-safe features, predictions, and audit views
AUDIENCE	You, interviewers, and reviewers who need clear field-level reasoning

UFC Data Dictionary Explainer

This PDF explains all data documented in docs/data_dictionary.md: what each table/view/feature means, why it exists, and how it supports scraping, warehousing, feature engineering, modeling, prediction, dashboarding, or post-event validation.

How to Read This Document

- Meaning: plain-English interpretation of the field or feature.
- Why used: the role it plays in joins, validation, model features, leakage prevention, reporting, or dashboard display.
- Fields marked as computed features are derived from prior fight history rather than directly scraped source columns.
- Debut prior features are included because the data dictionary documents them, but they are computed at training/inference time and are not stored in the database.

Data Flow

Source pages -> CSV snapshots -> normalized warehouse tables
-> fighter_snapshots -> bout_features
-> model predictions -> dashboard and post-event audit views

events

Event data anchors every fight in time and place. It is used for temporal splits, leakage-safe cutoffs, upcoming-vs-completed logic, dashboard filters, and post-event review.

event_id (type: uuid, nullable: PK)

Meaning: Unique event identifier from source
Why used: Used as a stable key for joins, traceability, and avoiding name-based ambiguity.

event_name (type: text, nullable: NOT NULL)

Meaning: e.g. "UFC 300: Pereira vs. Hill"
Why used: Used for human-readable display and dashboard/report labels.

event_date (type: date, nullable: NOT NULL)

Meaning: Scheduled date of the event
Why used: Used for temporal ordering, leakage-safe cutoffs, and event filtering.

city (type: text, nullable: yes)

Meaning: Venue city
Why used: Used for event context, filtering, and presentation.

state (type: text, nullable: yes)

Meaning: Venue state/province
Why used: Used for event context, filtering, and presentation.

country (type: text, nullable: yes)

Meaning: Venue country
Why used: Used for event context, filtering, and presentation.

event_status (type: text, nullable: yes)

Meaning: completed` or `upcoming
Why used: Used to separate upcoming data from completed historical data.

source_url (type: text, nullable: NOT NULL)

Meaning: ufcstats.com event page URL
Why used: Used for auditability so the record can be traced back to its source or artifact.

scraped_at (type: timestamptz, nullable: yes)

Meaning: When the record was last scraped
Why used: Used for freshness checks, reproducibility, and historical audit trails.

fighters

Fighter profile data stores relatively stable biographical and physical attributes. These fields support identity joins, age/reach/stance features, and dashboard fighter cards.

fighter_id (type: uuid, nullable: PK)

Meaning: Unique fighter identifier from source

Why used: Used as a stable key for joins, traceability, and avoiding name-based ambiguity.

full_name (type: text, nullable: NOT NULL)

Meaning: Full display name

Why used: Used for human-readable display and dashboard/report labels.

first_name (type: text, nullable: yes)

Meaning: First name

Why used: Used for human-readable display and dashboard/report labels.

last_name (type: text, nullable: yes)

Meaning: Last name

Why used: Used for human-readable display and dashboard/report labels.

nickname (type: text, nullable: yes)

Meaning: Fighter nickname

Why used: Used for human-readable display and dashboard/report labels.

height_cm (type: numeric(5,2), nullable: yes)

Meaning: Height in centimeters

Why used: Used as physical or demographic signal that can influence matchup advantages.

weight_lbs (type: numeric(5,1), nullable: yes)

Meaning: Weight in pounds

Why used: Used as physical or demographic signal that can influence matchup advantages.

reach_cm (type: numeric(5,2), nullable: yes)

Meaning: Reach in centimeters

Why used: Used as physical or demographic signal that can influence matchup advantages.

stance (type: text, nullable: yes)

Meaning: Fighting stance (see enums)

Why used: Used as physical or demographic signal that can influence matchup advantages.

dob (type: date, nullable: yes)

Meaning: Date of birth

Why used: Used as physical or demographic signal that can influence matchup advantages.

source_url (type: text, nullable: NOT NULL)

Meaning: ufcstats.com fighter page URL

Why used: Used for auditability so the record can be traced back to its source or artifact.

scraped_at (type: timestamptz, nullable: yes)

Meaning: When the record was last scraped

Why used: Used for freshness checks, reproducibility, and historical audit trails.

fight

Fight records are the central fact table. They connect events to two fighters, store the label/outcome, and provide matchup metadata such as weight class, title status, scheduled rounds, and finish details.

fight_id (type: uuid, nullable: PK)

Meaning: Unique fight identifier from source

Why used: Used as a stable key for joins, traceability, and avoiding name-based ambiguity.

event_id (type: uuid, nullable: NOT NULL)

Meaning: FK to `events.event\_id`

Why used: Used as a stable key for joins, traceability, and avoiding name-based ambiguity.

fighter_1_id (type: uuid, nullable: NOT NULL)

Meaning: FK to `fighters.fighter\_id` (deferred)

Why used: Used as a stable key for joins, traceability, and avoiding name-based ambiguity.

fighter_2_id (type: uuid, nullable: NOT NULL)

Meaning: FK to `fighters.fighter\_id` (deferred)

Why used: Used as a stable key for joins, traceability, and avoiding name-based ambiguity.

winner_fighter_id (type: uuid, nullable: yes)

Meaning: FK to `fighters.fighter\_id`. NULL for draws, NC, upcoming

Why used: Used as a stable key for joins, traceability, and avoiding name-based ambiguity.

result_type (type: text, nullable: NOT NULL)

Meaning: Fight outcome type (see enums)

Why used: Used because it carries business, modeling, validation, or dashboard context for the fight prediction system.

weight_class (type: text, nullable: yes)

Meaning: Weight division (see enums). NULL for early tournament bouts

Why used: Used as physical or demographic signal that can influence matchup advantages.

is_title_fight (type: boolean, nullable: NOT NULL)

Meaning: Whether this was a title fight

Why used: Used as matchup context because divisions, title fights, and 5-round fights behave differently.

is_interim_title (type: boolean, nullable: NOT NULL)

Meaning: Whether this was an interim title fight

Why used: Used as matchup context because divisions, title fights, and 5-round fights behave differently.

scheduled_rounds (type: smallint, nullable: yes)

Meaning: Number of scheduled rounds (3 or 5)

Why used: Used to understand fight outcome type, duration, and style of victory or loss.

finish_method (type: text, nullable: yes)

Meaning: How the fight ended (see enums)

Why used: Used to understand fight outcome type, duration, and style of victory or loss.

finish_detail (type: text, nullable: yes)

Meaning: Secondary method detail (free text)

Why used: Used to understand fight outcome type, duration, and style of victory or loss.

finish_round (type: smallint, nullable: yes)

Meaning: Round in which the fight ended

Why used: Used to understand fight outcome type, duration, and style of victory or loss.

finish_time_seconds (type: smallint, nullable: yes)

Meaning: Time in the finish round (seconds)

Why used: Used to understand fight outcome type, duration, and style of victory or loss.

referee (type: text, nullable: yes)

Meaning: Referee name

Why used: Used because it carries business, modeling, validation, or dashboard context for the fight prediction system.

source_url (type: text, nullable: NOT NULL)

Meaning: ufcstats.com fight page URL

Why used: Used for auditability so the record can be traced back to its source or artifact.

scraped_at (type: timestampz, nullable: yes)

Meaning: When the record was last scraped

Why used: Used for freshness checks, reproducibility, and historical audit trails.

fight_stats_aggregate

Aggregate fight statistics describe each fighter's total performance in a bout. They are used to build career, rolling, decay, striking, grappling, pace, damage, and control features.

fight_stat_id (type: uuid, nullable: PK)

Meaning: Unique stat row identifier
Why used: Used as a stable key for joins, traceability, and avoiding name-based ambiguity.

fight_id (type: uuid, nullable: NOT NULL)

Meaning: FK to `fights.fight\_id`
Why used: Used as a stable key for joins, traceability, and avoiding name-based ambiguity.

fighter_id (type: uuid, nullable: NOT NULL)

Meaning: FK to `fighters.fighter\_id`
Why used: Used as a stable key for joins, traceability, and avoiding name-based ambiguity.

knockdowns (type: smallint, nullable: NOT NULL)

Meaning: Total knockdowns scored
Why used: Used to measure striking volume, efficiency, damage, and defense.

total_strikes_landed (type: smallint, nullable: NOT NULL)

Meaning: All strikes landed
Why used: Used to measure striking volume, efficiency, damage, and defense.

total_strikes_attempted (type: smallint, nullable: NOT NULL)

Meaning: All strikes attempted
Why used: Used to measure striking volume, efficiency, damage, and defense.

sig_strikes_landed (type: smallint, nullable: NOT NULL)

Meaning: Significant strikes landed
Why used: Used to measure striking volume, efficiency, damage, and defense.

sig_strikes_attempted (type: smallint, nullable: NOT NULL)

Meaning: Significant strikes attempted
Why used: Used to measure striking volume, efficiency, damage, and defense.

sig_strikes_head_landed (type: smallint, nullable: NOT NULL)

Meaning: Sig strikes to head landed
Why used: Used to measure striking volume, efficiency, damage, and defense.

sig_strikes_head_attempted (type: smallint, nullable: NOT NULL)

Meaning: Sig strikes to head attempted
Why used: Used to measure striking volume, efficiency, damage, and defense.

sig_strikes_body_landed (type: smallint, nullable: NOT NULL)

Meaning: Sig strikes to body landed
Why used: Used to measure striking volume, efficiency, damage, and defense.

sig_strikes_body_attempted (type: smallint, nullable: NOT NULL)

Meaning: Sig strikes to body attempted
Why used: Used to measure striking volume, efficiency, damage, and defense.

sig_strikes_leg_landed (type: smallint, nullable: NOT NULL)

Meaning: Sig strikes to legs landed
Why used: Used to measure striking volume, efficiency, damage, and defense.

sig_strikes_leg_attempted (type: smallint, nullable: NOT NULL)

Meaning: Sig strikes to legs attempted
Why used: Used to measure striking volume, efficiency, damage, and defense.

sig_strikes_distance_landed (type: smallint, nullable: NOT NULL)

Meaning: Sig strikes at distance landed
Why used: Used as physical or demographic signal that can influence matchup advantages.

sig_strikes_distance_attempted (type: smallint, nullable: NOT NULL)

Meaning: Sig strikes at distance attempted

Why used: Used as physical or demographic signal that can influence matchup advantages.

sig_strikes_clinch_landed (type: smallint, nullable: NOT NULL)

Meaning: Sig strikes in clinch landed

Why used: Used to measure striking volume, efficiency, damage, and defense.

sig_strikes_clinch_attempted (type: smallint, nullable: NOT NULL)

Meaning: Sig strikes in clinch attempted

Why used: Used to measure striking volume, efficiency, damage, and defense.

sig_strikes_ground_landed (type: smallint, nullable: NOT NULL)

Meaning: Sig strikes on ground landed

Why used: Used to measure striking volume, efficiency, damage, and defense.

sig_strikes_ground_attempted (type: smallint, nullable: NOT NULL)

Meaning: Sig strikes on ground attempted

Why used: Used to measure striking volume, efficiency, damage, and defense.

takedowns_landed (type: smallint, nullable: NOT NULL)

Meaning: Successful takedowns

Why used: Used to measure grappling, wrestling, control, and submission threat.

takedowns_attempted (type: smallint, nullable: NOT NULL)

Meaning: Takedown attempts

Why used: Used to measure grappling, wrestling, control, and submission threat.

control_time_seconds (type: smallint, nullable: NOT NULL)

Meaning: Ground control time in seconds

Why used: Used to measure grappling, wrestling, control, and submission threat.

submissions_attempted (type: smallint, nullable: NOT NULL)

Meaning: Submission attempts

Why used: Used to measure grappling, wrestling, control, and submission threat.

reversals (type: smallint, nullable: NOT NULL)

Meaning: Reversals

Why used: Used to measure grappling, wrestling, control, and submission threat.

source_url (type: text, nullable: yes)

Meaning: Source page URL

Why used: Used for auditability so the record can be traced back to its source or artifact.

scraped_at (type: timestamp, nullable: yes)

Meaning: When scraped

Why used: Used for freshness checks, reproducibility, and historical audit trails.

fight_stats_by_round

Round-level statistics preserve the same performance signals at a finer grain. They are useful for deeper analysis, validation against aggregate totals, and future round-aware features.

fight_stat_by_round_id (type: uuid, nullable: PK)

Meaning: Unique stat row identifier

Why used: Used as a stable key for joins, traceability, and avoiding name-based ambiguity.

fight_id (type: uuid, nullable: NOT NULL)

Meaning: FK to `fights.fight\_id`

Why used: Used as a stable key for joins, traceability, and avoiding name-based ambiguity.

fighter_id (type: uuid, nullable: NOT NULL)

Meaning: FK to `fighters.fighter\_id`

Why used: Used as a stable key for joins, traceability, and avoiding name-based ambiguity.

round (type: smallint, nullable: NOT NULL)

Meaning: Round number (1-5)

Why used: Used to understand fight outcome type, duration, and style of victory or loss.

(all stat columns)

Meaning: Same as `fight\_stats\_aggregate`

Why used: Used because it carries business, modeling, validation, or dashboard context for the fight prediction system.

fighter_snapshots: Identifiers and metadata

These columns keep rows traceable, reproducible, and versioned.

fighter_id

Meaning: FK to `fighters.fighter\_id`

Why used: Used as a stable key for joins, traceability, and avoiding name-based ambiguity.

fight_id

Meaning: FK to `fights.fight\_id`

Why used: Used as a stable key for joins, traceability, and avoiding name-based ambiguity.

as_of_date

Meaning: Event date used as the feature cutoff

Why used: Used for temporal ordering, leakage-safe cutoffs, and event filtering.

feature_version

Meaning: Feature schema version

Why used: Used as part of this feature group to improve model input quality and interpretability.

computed_at

Meaning: Timestamp when the feature row was computed

Why used: Used for freshness checks, reproducibility, and historical audit trails.

fighter_snapshots: Career aggregates

Career aggregate features summarize the full prior UFC history available before the bout.

career_fights

Meaning: Total fights before this bout

Why used: Used as part of this feature group to improve model input quality and interpretability.

career_wins

Meaning: Total wins

Why used: Used as part of this feature group to improve model input quality and interpretability.

career_losses

Meaning: Total losses

Why used: Used as part of this feature group to improve model input quality and interpretability.

career_draws

Meaning: Total draws

Why used: Used as part of this feature group to improve model input quality and interpretability.

career_nc

Meaning: Total no-contests

Why used: Used as part of this feature group to improve model input quality and interpretability.

career_win_rate

Meaning: wins / fights

Why used: Used as normalized comparative signal that is easier for models to learn from than raw counts alone.

career_finish_rate

Meaning: (KO/TKO + sub wins) / wins

Why used: Used to understand fight outcome type, duration, and style of victory or loss.

career_ko_tko_wins

Meaning: KO/TKO victories

Why used: Used as part of this feature group to improve model input quality and interpretability.

career_sub_wins

Meaning: Submission victories

Why used: Used as part of this feature group to improve model input quality and interpretability.

career_dec_wins

Meaning: Decision victories

Why used: Used as part of this feature group to improve model input quality and interpretability.

career_ko_tko_losses

Meaning: KO/TKO losses

Why used: Used as part of this feature group to improve model input quality and interpretability.

career_sub_losses

Meaning: Submission losses

Why used: Used as part of this feature group to improve model input quality and interpretability.

career_title_fights

Meaning: Title fights competed in

Why used: Used as matchup context because divisions, title fights, and 5-round fights behave differently.

career_title_wins

Meaning: Title fights won

Why used: Used as matchup context because divisions, title fights, and 5-round fights behave differently.

career_minutes

Meaning: Total cage time in minutes

Why used: Used as part of this feature group to improve model input quality and interpretability.

career_sig_strikes_landed_pm

Meaning: Career sig strikes landed per minute

Why used: Used to measure striking volume, efficiency, damage, and defense.

career_sig_strikes_absorbed_pm

Meaning: Career sig strikes absorbed per minute

Why used: Used to measure striking volume, efficiency, damage, and defense.

career_sig_strike_accuracy

Meaning: Career sig strikes landed / attempted

Why used: Used to measure striking volume, efficiency, damage, and defense.

career_sig_strike_defense

Meaning: $1 - (\text{opponent sig strikes landed} / \text{attempted})$

Why used: Used to measure striking volume, efficiency, damage, and defense.

career_takedown_accuracy

Meaning: Career takedowns landed / attempted

Why used: Used to measure grappling, wrestling, control, and submission threat.

career_takedown_defense

Meaning: 1 - (opponent takedowns landed / attempted)

Why used: Used to measure grappling, wrestling, control, and submission threat.

career_sub_attempts_pm

Meaning: Submission attempts per 15 minutes

Why used: Used as part of this feature group to improve model input quality and interpretability.

career_control_rate

Meaning: Control time (seconds) per fight

Why used: Used to measure grappling, wrestling, control, and submission threat.

career_knockdowns_pm

Meaning: Knockdowns per fight

Why used: Used to measure striking volume, efficiency, damage, and defense.

fighter_snapshots: Rolling windows (last 1, 3, 5 fights)

Rolling windows capture recent form and reduce the influence of old fights.

win_rate

Meaning: Wins / N

Why used: Used as normalized comparative signal that is easier for models to learn from than raw counts alone.

finish_rate

Meaning: Finish wins / wins in window

Why used: Used to understand fight outcome type, duration, and style of victory or loss.

sig_strikes_landed_pm

Meaning: Sig strikes landed per minute

Why used: Used to measure striking volume, efficiency, damage, and defense.

sig_strikes_absorbed_pm

Meaning: Sig strikes absorbed per minute

Why used: Used to measure striking volume, efficiency, damage, and defense.

sig_strike_accuracy

Meaning: Sig strike accuracy

Why used: Used to measure striking volume, efficiency, damage, and defense.

sig_strike_defense

Meaning: Sig strike defense

Why used: Used to measure striking volume, efficiency, damage, and defense.

takedown_landed_pm

Meaning: Takedowns landed per minute

Why used: Used to measure grappling, wrestling, control, and submission threat.

takedown_accuracy

Meaning: Takedown accuracy

Why used: Used to measure grappling, wrestling, control, and submission threat.

takedown_defense

Meaning: Takedown defense

Why used: Used to measure grappling, wrestling, control, and submission threat.

control_rate

Meaning: Control time per fight

Why used: Used to measure grappling, wrestling, control, and submission threat.

knockdowns_pm

Meaning: Knockdowns per fight

Why used: Used to measure striking volume, efficiency, damage, and defense.

knockdowns_absorbed_pm

Meaning: Knockdowns absorbed per fight

Why used: Used to measure striking volume, efficiency, damage, and defense.

sub_attempts_pm

Meaning: Submission attempts per minute

Why used: Used as part of this feature group to improve model input quality and interpretability.

avg_fight_time

Meaning: Average fight duration in minutes

Why used: Used to understand fight outcome type, duration, and style of victory or loss.

streak

Meaning: Current win/loss streak over the window

Why used: Used to capture recent form, momentum, recency, or volatility.

fighter_snapshots: Exponentially decayed metrics (alpha = 0.85)

Decayed metrics keep all prior history but weight recent fights more heavily.

sig_strikes_landed_pm_decay

Meaning: Recency-weighted sig strike rate

Why used: Used to measure striking volume, efficiency, damage, and defense.

sig_strikes_absorbed_pm_decay

Meaning: Recency-weighted sig strikes absorbed rate

Why used: Used to measure striking volume, efficiency, damage, and defense.

sig_strike_accuracy_decay

Meaning: Recency-weighted sig strike accuracy

Why used: Used to measure striking volume, efficiency, damage, and defense.

sig_strike_defense_decay

Meaning: Recency-weighted sig strike defense

Why used: Used to measure striking volume, efficiency, damage, and defense.

takedown_landed_pm_decay

Meaning: Recency-weighted takedown rate

Why used: Used to measure grappling, wrestling, control, and submission threat.

takedown_accuracy_decay

Meaning: Recency-weighted takedown accuracy

Why used: Used to measure grappling, wrestling, control, and submission threat.

takedown_defense_decay

Meaning: Recency-weighted takedown defense

Why used: Used to measure grappling, wrestling, control, and submission threat.

control_rate_decay

Meaning: Recency-weighted control time

Why used: Used to measure grappling, wrestling, control, and submission threat.

knockdowns_pm_decay

Meaning: Recency-weighted knockdown rate

Why used: Used to measure striking volume, efficiency, damage, and defense.

win_rate_decay

Meaning: Recency-weighted win rate

Why used: Used to capture recent form, momentum, recency, or volatility.

fighter_snapshots: v2 trend features (Phase 5)

Trend and volatility features describe whether recent performance is improving, declining, or unstable.

slope_sig_strikes_last5

Meaning: Linear trend slope of sig strike rate over last 5 fights

Why used: Used to measure striking volume, efficiency, damage, and defense.

slope_td_accuracy_last5

Meaning: Linear trend slope of takedown accuracy over last 5 fights

Why used: Used to capture recent form, momentum, recency, or volatility.

slope_control_rate_last5

Meaning: Linear trend slope of control rate over last 5 fights

Why used: Used to measure grappling, wrestling, control, and submission threat.

std_sig_strikes_last5

Meaning: Standard deviation of sig strike rate over last 5 fights

Why used: Used to measure striking volume, efficiency, damage, and defense.

std_td_accuracy_last5

Meaning: Standard deviation of takedown accuracy over last 5 fights

Why used: Used to capture recent form, momentum, recency, or volatility.

fights_per_year_last3

Meaning: Fight frequency: fights per year over last 3 fights

Why used: Used to capture recent form, momentum, recency, or volatility.

fighter_snapshots: Physical, demographic, activity

Physical and activity fields capture attributes and layoff effects that raw fight records do not express directly.

age

Meaning: Age at fight date (years)

Why used: Used as physical or demographic signal that can influence matchup advantages.

age_squared

Meaning: Age squared (captures non-linear aging effects)

Why used: Used as physical or demographic signal that can influence matchup advantages.

height_cm

Meaning: Fighter height

Why used: Used as physical or demographic signal that can influence matchup advantages.

reach_cm

Meaning: Fighter reach

Why used: Used as physical or demographic signal that can influence matchup advantages.

reach_to_height

Meaning: Reach / height ratio

Why used: Used as physical or demographic signal that can influence matchup advantages.

is_orthodox

Meaning: Orthodox stance flag

Why used: Used as part of this feature group to improve model input quality and interpretability.

is_southpaw

Meaning: Southpaw stance flag

Why used: Used as part of this feature group to improve model input quality and interpretability.

days_since_last_fight

Meaning: Days since previous fight

Why used: Used to capture recent form, momentum, recency, or volatility.

is_long_layoff

Meaning: True if > 365 days since last fight

Why used: Used as part of this feature group to improve model input quality and interpretability.

is_short_notice

Meaning: Short-notice fight flag

Why used: Used as part of this feature group to improve model input quality and interpretability.

is_debut

Meaning: True if this is the fighter's first fight in the dataset

Why used: Used as part of this feature group to improve model input quality and interpretability.

age_missing

Meaning: True if DOB is unknown

Why used: Used as physical or demographic signal that can influence matchup advantages.

height_reach_missing

Meaning: True if height or reach is unknown

Why used: Used as physical or demographic signal that can influence matchup advantages.

fighter_snapshots: Elo and opponent strength

Elo and opponent-strength features estimate fighter quality and schedule difficulty sequentially.

elo_rating

Meaning: Pre-fight Elo rating (starts at 1500)

Why used: Used as a compact sequential estimate of fighter strength and opponent quality.

elo_opponent

Meaning: Opponent's pre-fight Elo rating

Why used: Used as a compact sequential estimate of fighter strength and opponent quality.

elo_diff

Meaning: $\text{elo_rating} - \text{elo_opponent}$

Why used: Used as a compact sequential estimate of fighter strength and opponent quality.

opp_avg_elo

Meaning: Average Elo of prior opponents

Why used: Used as a compact sequential estimate of fighter strength and opponent quality.

opp_adj_sig_strike_accuracy

Meaning: Sig strike accuracy adjusted for opponent quality

Why used: Used to measure striking volume, efficiency, damage, and defense.

bout_features: Identifiers and metadata

These columns keep rows traceable, reproducible, and versioned.

fight_id

Meaning: PK, FK to `fights.fight\_id`

Why used: Used as a stable key for joins, traceability, and avoiding name-based ambiguity.

fighter_1_id

Meaning: FK to `fighters.fighter\_id`

Why used: Used as a stable key for joins, traceability, and avoiding name-based ambiguity.

fighter_2_id

Meaning: FK to `fighters.fighter\_id`

Why used: Used as a stable key for joins, traceability, and avoiding name-based ambiguity.

event_date

Meaning: Fight date

Why used: Used for temporal ordering, leakage-safe cutoffs, and event filtering.

weight_class

Meaning: Weight division

Why used: Used as physical or demographic signal that can influence matchup advantages.

is_title_fight

Meaning: Title fight flag

Why used: Used as matchup context because divisions, title fights, and 5-round fights behave differently.

scheduled_rounds

Meaning: 3 or 5

Why used: Used to understand fight outcome type, duration, and style of victory or loss.

label

Meaning: 1 = fighter_1 won, 0 = fighter_2 won, NULL = draw/NC

Why used: Used to train or evaluate the prediction model against actual outcomes.

feature_version

Meaning: 1 (v1) or 2 (v2)

Why used: Used as part of this feature group to improve model input quality and interpretability.

computed_at

Meaning: Timestamp when the feature row was computed

Why used: Used for freshness checks, reproducibility, and historical audit trails.

bout_features: Difference features (fighter_1 - fighter_2)

Difference features express direct matchup advantages for fighter_1 over fighter_2.

diff_elo

Meaning: Elo rating difference

Why used: Used as a compact sequential estimate of fighter strength and opponent quality.

diff_career_wins

Meaning: Win count difference

Why used: Used as normalized comparative signal that is easier for models to learn from than raw counts alone.

diff_career_fights

Meaning: Fight count difference

Why used: Used as normalized comparative signal that is easier for models to learn from than raw counts alone.

diff_career_win_rate

Meaning: Win rate difference

Why used: Used as normalized comparative signal that is easier for models to learn from than raw counts alone.

diff_career_finish_rate

Meaning: Finish rate difference

Why used: Used to understand fight outcome type, duration, and style of victory or loss.

diff_career_sig_strikes_landed_pm

Meaning: Sig strike rate difference

Why used: Used to measure striking volume, efficiency, damage, and defense.

diff_career_sig_strike_accuracy

Meaning: Sig strike accuracy difference

Why used: Used to measure striking volume, efficiency, damage, and defense.

diff_career_takedown_accuracy

Meaning: Takedown accuracy difference

Why used: Used to measure grappling, wrestling, control, and submission threat.

diff_career_control_rate

Meaning: Control time per fight difference

Why used: Used to measure grappling, wrestling, control, and submission threat.

diff_age

Meaning: Age difference

Why used: Used as physical or demographic signal that can influence matchup advantages.

diff_height_cm

Meaning: Height difference

Why used: Used as physical or demographic signal that can influence matchup advantages.

diff_reach_cm

Meaning: Reach difference

Why used: Used as physical or demographic signal that can influence matchup advantages.

diff_days_since_last_fight

Meaning: Days since last fight difference

Why used: Used to capture recent form, momentum, recency, or volatility.

diff_win_rate_last3

Meaning: Rolling 3-fight win rate difference

Why used: Used to capture recent form, momentum, recency, or volatility.

diff_sig_strikes landed_pm_last3

Meaning: Rolling 3-fight sig strike rate difference

Why used: Used to measure striking volume, efficiency, damage, and defense.

diff_takedown_accuracy_last3

Meaning: Rolling 3-fight takedown accuracy difference

Why used: Used to measure grappling, wrestling, control, and submission threat.

diff_control_rate_last3

Meaning: Rolling 3-fight control rate difference

Why used: Used to measure grappling, wrestling, control, and submission threat.

diff_sig_strikes landed_pm_decay

Meaning: Decayed sig strike rate difference

Why used: Used to measure striking volume, efficiency, damage, and defense.

diff_win_rate_decay

Meaning: Decayed win rate difference

Why used: Used to capture recent form, momentum, recency, or volatility.

diff_opp_avg_elo

Meaning: Average opponent Elo difference

Why used: Used as a compact sequential estimate of fighter strength and opponent quality.

bout_features: v2 difference features (Phase 5)

These additions add more style, defense, experience, trend, and activity signal to the matchup row.

diff_career_ko_rate

Meaning: KO/TKO win rate difference

Why used: Used as normalized comparative signal that is easier for models to learn from than raw counts alone.

diff_career_sub_rate

Meaning: Submission win rate difference

Why used: Used as normalized comparative signal that is easier for models to learn from than raw counts alone.

diff_career_decision_rate

Meaning: Decision win rate difference

Why used: Used as normalized comparative signal that is easier for models to learn from than raw counts alone.

diff_career_sig_strikes_absorbed_pm

Meaning: Sig strikes absorbed rate difference

Why used: Used to measure striking volume, efficiency, damage, and defense.

diff_career_sig_strike_defense

Meaning: Sig strike defense difference

Why used: Used to measure striking volume, efficiency, damage, and defense.

diff_career_takedown_defense

Meaning: Takedown defense difference

Why used: Used to measure grappling, wrestling, control, and submission threat.

diff_title_fight_count

Meaning: Title fight experience difference

Why used: Used as matchup context because divisions, title fights, and 5-round fights behave differently.

diff_five_round_fights

Meaning: Five-round fight experience difference

Why used: Used to understand fight outcome type, duration, and style of victory or loss.

diff_reach_height_ratio

Meaning: Reach-to-height ratio difference

Why used: Used as physical or demographic signal that can influence matchup advantages.

diff_fights_per_year_last3

Meaning: Fight frequency difference

Why used: Used to capture recent form, momentum, recency, or volatility.

diff_slope_sig_strikes_last5

Meaning: Sig strike trend difference

Why used: Used to measure striking volume, efficiency, damage, and defense.

diff_slope_td_accuracy_last5

Meaning: Takedown accuracy trend difference

Why used: Used to capture recent form, momentum, recency, or volatility.

diff_slope_control_rate_last5

Meaning: Control rate trend difference

Why used: Used to measure grappling, wrestling, control, and submission threat.

diff_std_sig_strikes_last5

Meaning: Sig strike volatility difference

Why used: Used to measure striking volume, efficiency, damage, and defense.

diff_std_td_accuracy_last5

Meaning: Takedown accuracy volatility difference

Why used: Used to capture recent form, momentum, recency, or volatility.

bout_features: Ratio features ($\text{fighter}_1 / (\text{fighter}_1 + \text{fighter}_2)$)

Ratio features bound comparative quantities between 0 and 1 and reduce scale issues.

ratio_career_wins

Meaning: Win count ratio

Why used: Used as normalized comparative signal that is easier for models to learn from than raw counts alone.

ratio_career_fights

Meaning: Fight count ratio

Why used: Used as normalized comparative signal that is easier for models to learn from than raw counts alone.

ratio_career_sig_strikes_landed_pm

Meaning: Sig strike rate ratio

Why used: Used to measure striking volume, efficiency, damage, and defense.

ratio_career_control_rate

Meaning: Control rate ratio

Why used: Used to measure grappling, wrestling, control, and submission threat.

ratio_elo

Meaning: Elo rating ratio

Why used: Used as a compact sequential estimate of fighter strength and opponent quality.

bout_features: Matchup and metadata features

These fields represent matchup structure and categorical context that can change model behavior.

is_orthodox_vs_southpaw

Meaning: True if one orthodox, one southpaw

Why used: Used as part of this feature group to improve model input quality and interpretability.

both_debuting

Meaning: True if both fighters debuting

Why used: Used as part of this feature group to improve model input quality and interpretability.

f1_is_southpaw

Meaning: Fighter 1 southpaw flag

Why used: Used as part of this feature group to improve model input quality and interpretability.

f2_is_southpaw

Meaning: Fighter 2 southpaw flag

Why used: Used as part of this feature group to improve model input quality and interpretability.

weight_class_rank

Meaning: Ordinal weight class encoding (1=strawweight to 9=heavyweight)

Why used: Used as physical or demographic signal that can influence matchup advantages.

bout_features: Debut prior features (computed at training/inference time, not in

Debut priors provide limited signal for sparse-history fighters without pretending there is rich prior performance data.

debut_prior_win_prob_f1

Meaning: Fixed at 0.5 for debut bouts, NaN otherwise

Why used: Used to expose model output, confidence, and uncertainty to users and audits.

debut_reach_adv

Meaning: Reach difference as z-score by weight class, NaN for non-debut

Why used: Used as physical or demographic signal that can influence matchup advantages.

debut_height_adv

Meaning: Height difference as z-score by weight class, NaN for non-debut

Why used: Used as physical or demographic signal that can influence matchup advantages.

predictions

Predictions store each scoring run, not just the latest result. Keeping scored_at in the primary key makes post-event audits honest because the system can prove a prediction existed before event day.

fight_id (type: uuid, nullable: PK, NOT NULL)

Meaning: FK to `fights.fight\_id`

Why used: Used as a stable key for joins, traceability, and avoiding name-based ambiguity.

event_date (type: date, nullable: yes)

Meaning: Date of the event being scored

Why used: Used for temporal ordering, leakage-safe cutoffs, and event filtering.

fighter_1_id (type: uuid, nullable: NOT NULL)

Meaning: FK to `fighters.fighter\_id`

Why used: Used as a stable key for joins, traceability, and avoiding name-based ambiguity.

fighter_2_id (type: uuid, nullable: NOT NULL)

Meaning: FK to `fighters.fighter\_id`

Why used: Used as a stable key for joins, traceability, and avoiding name-based ambiguity.

fighter_1_name (type: text, nullable: yes)

Meaning: Fighter 1 display name copied for dashboard/export convenience

Why used: Used for human-readable display and dashboard/report labels.

fighter_2_name (type: text, nullable: yes)

Meaning: Fighter 2 display name copied for dashboard/export convenience

Why used: Used for human-readable display and dashboard/report labels.

weight_class (type: text, nullable: yes)

Meaning: Bout weight division

Why used: Used as physical or demographic signal that can influence matchup advantages.

predicted_prob_f1 (type: numeric(6,4), nullable: yes)

Meaning: Raw model probability that fighter_1 wins

Why used: Used to expose model output, confidence, and uncertainty to users and audits.

calibrated_prob_f1 (type: numeric(6,4), nullable: yes)

Meaning: Calibrated probability after Platt scaling

Why used: Used to expose model output, confidence, and uncertainty to users and audits.

confidence_tier (type: text, nullable: yes)

Meaning: high`, `medium`, or `toss-up

Why used: Used to expose model output, confidence, and uncertainty to users and audits.

is_uncertain (type: boolean, nullable: yes)

Meaning: True if the fight is in the model's uncertain probability band

Why used: Used to expose model output, confidence, and uncertainty to users and audits.

model_name (type: text, nullable: NOT NULL)

Meaning: Name of the model used for scoring

Why used: Used for human-readable display and dashboard/report labels.

model_artifact (type: text, nullable: yes)

Meaning: Path to the model artifact used for scoring

Why used: Used for auditability so the record can be traced back to its source or artifact.

scored_at (type: timestamptz, nullable: PK, NOT NULL)

Meaning: Timestamp for the scoring run

Why used: Used for freshness checks, reproducibility, and historical audit trails.

fighter_career_summary

This dashboard view turns the latest fighter snapshots into an easy fighter profile summary with record, style, recent form, activity, and latest-fight context.

fighter_id

Meaning: Fighter identifier

Why used: Used as a stable key for joins, traceability, and avoiding name-based ambiguity.

full_name

Meaning: Fighter display name

Why used: Used for human-readable display and dashboard/report labels.

first_name

Meaning: First name

Why used: Used for human-readable display and dashboard/report labels.

last_name

Meaning: Last name

Why used: Used for human-readable display and dashboard/report labels.

nickname

Meaning: Fighter nickname

Why used: Used for human-readable display and dashboard/report labels.

height_cm

Meaning: Height in centimeters

Why used: Used as physical or demographic signal that can influence matchup advantages.

weight_lbs

Meaning: Weight in pounds

Why used: Used as physical or demographic signal that can influence matchup advantages.

reach_cm

Meaning: Reach in centimeters

Why used: Used as physical or demographic signal that can influence matchup advantages.

stance

Meaning: Fighting stance

Why used: Used as physical or demographic signal that can influence matchup advantages.

dob

Meaning: Date of birth

Why used: Used as physical or demographic signal that can influence matchup advantages.

total_fights

Meaning: Career fights including latest fight

Why used: Used because it carries business, modeling, validation, or dashboard context for the fight prediction system.

total_wins

Meaning: Career wins including latest fight

Why used: Used because it carries business, modeling, validation, or dashboard context for the fight prediction system.

total_losses

Meaning: Career losses including latest fight

Why used: Used because it carries business, modeling, validation, or dashboard context for the fight prediction system.

total_draws

Meaning: Career draws including latest fight

Why used: Used because it carries business, modeling, validation, or dashboard context for the fight prediction system.

career_nc

Meaning: Career no-contests

Why used: Used because it carries business, modeling, validation, or dashboard context for the fight prediction system.

win_rate

Meaning: $\text{total_wins} / \text{total_fights}$

Why used: Used as normalized comparative signal that is easier for models to learn from than raw counts alone.

finish_rate

Meaning: Career finish rate from latest snapshot

Why used: Used to understand fight outcome type, duration, and style of victory or loss.

career_ko_tko_wins

Meaning: KO/TKO wins

Why used: Used because it carries business, modeling, validation, or dashboard context for the fight prediction system.

career_sub_wins

Meaning: Submission wins

Why used: Used because it carries business, modeling, validation, or dashboard context for the fight prediction system.

career_dec_wins

Meaning: Decision wins

Why used: Used because it carries business, modeling, validation, or dashboard context for the fight prediction system.

career_ko_tko_losses

Meaning: KO/TKO losses

Why used: Used because it carries business, modeling, validation, or dashboard context for the fight prediction system.

career_sub_losses

Meaning: Submission losses

Why used: Used because it carries business, modeling, validation, or dashboard context for the fight prediction system.

career_title_fights

Meaning: Title fights competed in

Why used: Used as matchup context because divisions, title fights, and 5-round fights behave differently.

career_title_wins

Meaning: Title fights won

Why used: Used as matchup context because divisions, title fights, and 5-round fights behave differently.

total_minutes

Meaning: Career cage time in minutes

Why used: Used because it carries business, modeling, validation, or dashboard context for the fight prediction system.

sig_strikes_landed_pm

Meaning: Career significant strikes landed per minute

Why used: Used to measure striking volume, efficiency, damage, and defense.

sig_strikes_absorbed_pm

Meaning: Career significant strikes absorbed per minute

Why used: Used to measure striking volume, efficiency, damage, and defense.

sig_strike_accuracy

Meaning: Career significant strike accuracy

Why used: Used to measure striking volume, efficiency, damage, and defense.

sig_strike_defense

Meaning: Career significant strike defense

Why used: Used to measure striking volume, efficiency, damage, and defense.

takedown_accuracy

Meaning: Career takedown accuracy

Why used: Used to measure grappling, wrestling, control, and submission threat.

takedown_defense

Meaning: Career takedown defense

Why used: Used to measure grappling, wrestling, control, and submission threat.

control_rate

Meaning: Career control rate

Why used: Used to measure grappling, wrestling, control, and submission threat.

knockdowns_pm

Meaning: Career knockdowns per minute/fight metric from snapshots

Why used: Used to measure striking volume, efficiency, damage, and defense.

elo_rating

Meaning: Latest Elo rating

Why used: Used as a compact sequential estimate of fighter strength and opponent quality.

win_rate_last3

Meaning: Rolling 3-fight win rate

Why used: Used to capture recent form, momentum, recency, or volatility.

win_rate_last5

Meaning: Rolling 5-fight win rate

Why used: Used to capture recent form, momentum, recency, or volatility.

sig_strikes_landed_pm_last3

Meaning: Rolling 3-fight significant strike rate

Why used: Used to measure striking volume, efficiency, damage, and defense.

sig_strikes_landed_pm_last5

Meaning: Rolling 5-fight significant strike rate

Why used: Used to measure striking volume, efficiency, damage, and defense.

takedown_accuracy_last3

Meaning: Rolling 3-fight takedown accuracy

Why used: Used to measure grappling, wrestling, control, and submission threat.

takedown_accuracy_last5

Meaning: Rolling 5-fight takedown accuracy

Why used: Used to measure grappling, wrestling, control, and submission threat.

control_rate_last3

Meaning: Rolling 3-fight control rate

Why used: Used to measure grappling, wrestling, control, and submission threat.

control_rate_last5

Meaning: Rolling 5-fight control rate

Why used: Used to measure grappling, wrestling, control, and submission threat.

streak_last5

Meaning: Recent streak over last 5 fights

Why used: Used to capture recent form, momentum, recency, or volatility.

age

Meaning: Latest snapshot age

Why used: Used as physical or demographic signal that can influence matchup advantages.

days_since_last_fight

Meaning: Days since previous fight as of latest snapshot

Why used: Used to capture recent form, momentum, recency, or volatility.

last_fight_outcome

Meaning: Latest fight outcome from the fighter perspective

Why used: Used to capture recent form, momentum, recency, or volatility.

last_fight_method

Meaning: Latest fight finish method

Why used: Used to capture recent form, momentum, recency, or volatility.

last_event_name

Meaning: Latest event name

Why used: Used for human-readable display and dashboard/report labels.

last_fight_date

Meaning: Latest fight date

Why used: Used for temporal ordering, leakage-safe cutoffs, and event filtering.

result_type (fights)

win

Meaning: One fighter won (winner_fighter_id is set)

Why used: Used because it carries business, modeling, validation, or dashboard context for the fight prediction system.

draw

Meaning: Fight ended in a draw

Why used: Used because it carries business, modeling, validation, or dashboard context for the fight prediction system.

nc

Meaning: No contest

Why used: Used because it carries business, modeling, validation, or dashboard context for the fight prediction system.

upcoming

Meaning: Fight has not yet occurred

Why used: Used because it carries business, modeling, validation, or dashboard context for the fight prediction system.

event_status (events)

completed

Meaning: Event has taken place

Why used: Used because it carries business, modeling, validation, or dashboard context for the fight prediction system.

upcoming

Meaning: Event is scheduled but has not occurred

Why used: Used because it carries business, modeling, validation, or dashboard context for the fight prediction system.

weight_class (fights, bout_features)

strawweight

Meaning: 1

Why used: Used as physical or demographic signal that can influence matchup advantages.

flyweight

Meaning: 2

Why used: Used as physical or demographic signal that can influence matchup advantages.

bantamweight

Meaning: 3

Why used: Used as physical or demographic signal that can influence matchup advantages.

featherweight

Meaning: 4

Why used: Used as physical or demographic signal that can influence matchup advantages.

lightweight

Meaning: 5

Why used: Used as physical or demographic signal that can influence matchup advantages.

welterweight

Meaning: 6

Why used: Used as physical or demographic signal that can influence matchup advantages.

middleweight

Meaning: 7

Why used: Used as physical or demographic signal that can influence matchup advantages.

light_heavyweight

Meaning: 8

Why used: Used as physical or demographic signal that can influence matchup advantages.

heavyweight

Meaning: 9

Why used: Used as physical or demographic signal that can influence matchup advantages.

women_strawweight

Meaning: 1

Why used: Used as physical or demographic signal that can influence matchup advantages.

women_flyweight

Meaning: 2

Why used: Used as physical or demographic signal that can influence matchup advantages.

women_bantamweight

Meaning: 3

Why used: Used as physical or demographic signal that can influence matchup advantages.

women_featherweight

Meaning: 4

Why used: Used as physical or demographic signal that can influence matchup advantages.

catch_weight

Meaning: NULL

Why used: Used as physical or demographic signal that can influence matchup advantages.

open_weight

Meaning: NULL

Why used: Used as physical or demographic signal that can influence matchup advantages.

stance (fighters)

orthodox

Meaning: Right-handed stance (left foot forward)

Why used: Used because it carries business, modeling, validation, or dashboard context for the fight prediction system.

southpaw

Meaning: Left-handed stance (right foot forward)

Why used: Used because it carries business, modeling, validation, or dashboard context for the fight prediction system.

switch

Meaning: Switches between stances

Why used: Used because it carries business, modeling, validation, or dashboard context for the fight prediction system.

finish_method (fights)

decision

Meaning: Went to judges' scorecards

Why used: Used because it carries business, modeling, validation, or dashboard context for the fight prediction system.

ko_tko

Meaning: Knockout or technical knockout

Why used: Used because it carries business, modeling, validation, or dashboard context for the fight prediction system.

submission

Meaning: Submission finish

Why used: Used to measure grappling, wrestling, control, and submission threat.

doctor_stoppage

Meaning: Stopped by ringside doctor

Why used: Used as physical or demographic signal that can influence matchup advantages.

overturned

Meaning: Result overturned (e.g. failed drug test)

Why used: Used because it carries business, modeling, validation, or dashboard context for the fight prediction system.

could_not_continue

Meaning: Fighter could not continue (injury)

Why used: Used because it carries business, modeling, validation, or dashboard context for the fight prediction system.

dq

Meaning: Disqualification

Why used: Used because it carries business, modeling, validation, or dashboard context for the fight prediction system.

other

Meaning: Other finish method

Why used: Used because it carries business, modeling, validation, or dashboard context for the fight prediction system.

confidence_tier (predictions)

high

Meaning: Model is relatively confident

Why used: Used because it carries business, modeling, validation, or dashboard context for the fight prediction system.

medium

Meaning: Moderate confidence

Why used: Used because it carries business, modeling, validation, or dashboard context for the fight prediction system.

toss-up

Meaning: Near coin-flip, low signal

Why used: Used because it carries business, modeling, validation, or dashboard context for the fight prediction system.

Feature Set Versions

v1 (29 features)

The v1 feature set is the original model-ready matchup schema. It combines Elo, career record, core striking and grappling metrics, physical differences, recent form, decayed performance, opponent strength, and matchup metadata.

Why used: it created a strong baseline feature contract for Logistic Regression and early LightGBM models while remaining compact enough to inspect and debug.

v2 + debut priors (50 features)

The v2 feature set extends v1 with finish-style rates, defense, title/five-round experience, reach-height ratio, activity frequency, trend and volatility features, stance detail, weight-class rank, and debut-specific priors.

Why used: it gives boosted tree models richer signal about style, durability, matchup context, sparse-history fighters, and recent performance trajectory.

Relationships

The relationship graph explains how stored data connects across the project.

```
events 1—N fights
fighters 1—N fights (as fighter_1_id or fighter_2_id)
fights 1—2 fight_stats_aggregate (one per fighter)
fights 1—N fight_stats_by_round (one per fighter per round)
fights 1—2 fighter_snapshots (one per fighter)
fights 1—1 bout_features
fights 1—N predictions (one per scoring run)
predictions N—1 latest_predictions view source
predictions + fights + events N—1 current_event_predictions view source
predictions + fights + events N—1 pre_event_prediction_fights view source
pre_event_prediction_fights N—1 pre_event_prediction_events view source
fighter_snapshots N—1 fighter_career_summary view source
```

- events to fights: one card contains many bouts.
- fighters to fights: each bout references two fighters.
- fights to stats: each completed bout can have two aggregate stat rows and many round rows.
- fights to fighter_snapshots and bout_features: the feature layer is keyed to the fight being predicted.
- fights to predictions: the same fight may be scored multiple times across model runs.
- prediction views: SQL views turn raw prediction history into latest, current-event, and pre-event audit outputs.